Ch 13: Radio Buttons, List Boxes and Canvas Drawing

Application Program Development

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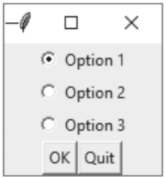
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Radio Buttons

Radio Buttons

Radio buttons normally appear in groups of two or more, and allow the user to select one of several possible options. Check buttons, which may appear alone or in groups, allow the user to make yes/no or on/off selections.



- **Radio button**: small circle that appears filled in when it is selected, and appears empty when it is deselected
  - Useful when you want the user to select one choice from several possible options
- **`Radiobutton` widgets**: created using `tkinter` module's `Radiobutton` class
  - `Radiobutton` widgets are mutually exclusive
  - Only one radio button in a container may be selected at any given time
- **`IntVar` class**: a `tkinter` module class that can be used along with `Radiobutton` widgets
  - Steps for use:
    - Associate group of `Radiobutton` widgets with the same `IntVar` object
    - Assign unique integer to each `Radiobutton`
    - When a `Radiobutton` widget is selected, its unique integer is stored in the `IntVar` object
  - Can be used to select a default radio button

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Using Callback Functions with Radiobuttons

- You can specify a callback function with `Radiobutton` widgets
  - Provide an argument `command=self.my_method` when creating the `Radiobutton` widget
  - The command will execute immediately when the radio button is selected
  - Replaces the need for a user to click OK or submit before determining which `Radiobutton` is selected

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Check Buttons

- **Check button**: small box with a label appearing next to it, check mark indicates when it is selected
  - User is allowed to select any or all of the check buttons that are displayed in a group
  - Not mutually exclusive
- **`Checkbutton` widgets**: created using `tkinter` module's `Checkbutton` class
  - Associate different `IntVar` object with each `Checkbutton` widget

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Listbox Widgets

- A `Listbox` widget displays a list of items and allows the user to select an item from the list.
  - Can be mutually exclusive or not
  - Easier to dynamically change list of options to select than a radio or check box
- Specify the size of a `Listbox`:
  - The `height` of a `Listbox` will determine number of lines it will display
  - The `width` of a `Listbox` is number of characters, so `width=30` has a listbox that can display 30 characters



Figure 2: An example of a tkinter Listbox widget (Gaddis, 2021, pg.691)

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Listbox Retrieving the Selected Item(s)

- The `curselection()` method returns a tuple containing the indexes of the items that are currently selected in the `Listbox`
  - If no item is selected, the tuple will be empty
  - If an item is selection and mode is `BROWSE` or `SINGLE` the tuple will contain only 1 element
  - If the mode is `EXTENDED` or `MULTIPLE` the tuple can contain multiple elements
- Once you have the index of the selected item, you can use the `get()` method to retrieve
- The `get()` method accepts an integer index as its argument and returns the item that is located at that index in the `Listbox`

Table 1: Listbox selection modes

Mode	Description
`tkinter.BROWSE`	The user can select one item at a time. If drags item under cursor is selected.
`tkinter.EXTENDED`	The user can select a group of adjacent items by clicking and dragging.
`tkinter.MULTIPLE`	Multiple items can be selected. Click selects / unselects items in list.
`tkinter.SINGLE`	The user can select one item at a time by clicking in the list.

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Listboxes and Callback Functions

- You can optionally bind a callback function to a `Listbox`
- When the user clicks an item in the `Listbox` the callback function is immediately executed
- Example callback function:
  - You must provide a parameter to accept an event object as an argument
  - To bind the `show_item` function to the `listbox` `widget:self.listbox.bind(<<ListboxSelect>>, self.show_item)‘`
  - Once done, the `show_item` function will execute whenever the user clicks an item in the `listbox` widget



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Adding Vertical and Horizontal Scrollbars to a `Listbox`

1. Create the outer frame to hold the inner frame and the horizontal scrollbar
2. Pack the outer frame
3. Create the inner frame to hold the `Listbox` and the vertical scrollbar
4. Pack the inner frame
5. Create a `Listbox` inside the inner frame
6. Pack the `Listbox` to the left side of the inner frame
7. Create a vertical scrollbar inside the inner frame
8. Pack the scrollbar ro the right side of the inner frame
9. Create the horizontal scrollbar inside the outer frame



Drawing Shapes with the Canvas Widget

Canvas Widget

The `Canvas` widget provides methods for drawing simple shapes, such as lines, rectangle, ovals, polygons, and so on.

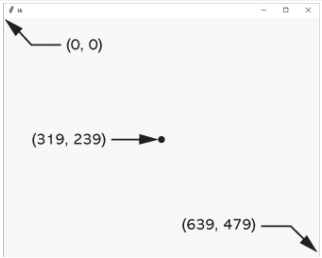


Figure 3: Various pixel locations in a 640 by 480 pixel sized window (Gaddis, 2021,

- The `Canvas` widget is a blank, rectangular area that allows you to draw simple 2D shapes
- You can use the `Canvas` widget's **screen coordinate system** to specify the location of your graphics
- The coordinates of the pixel in the upper-left corner of the screen are `(0, 0)`
  - The `X` coordinates increase from left to right
  - The `Y` coordinates increase from top to bottom
- The `Canvas` widget has numerous methods for drawing graphical shapes on the surface of the widget
  - `create_line()`
  - `create_rectangle()`
  - `create_oval()`
  - `create_arc()`
  - `create_polygon()`
  - `create_text()`

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Example of Drawing with the Canvas Widget

Look at the examples in `code/ch13`:

- `draw_line.py()`
- `draw_multi_line.py()`
- `draw_square.py()`
- `draw_ovals.py()`
- `draw_arc.py()`
- `draw_polygon().py`
- `draw_text().py`



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Summary

- A `Radiobutton` allows for one choice of several possible options (mutually exclusive)
  - Can use `IntVar` to get and select default for selected `Radiobutton`
- A `Checkbutton` allows for multiple choice among several possible options (not mutually exclusive)
- A `Listbox` displays textual list, and can be used for single or multiple choice
  - Can be more flexible than radio buttons or list boxes, can build dynamically
  - Adding scrollbars to `tkinter` elements can be a bit tricky...
- The `Canvas` widget provides a canvas that you can draw arbitrary geometric shapes on
  - Methods with support for lines, rectangles, ovals, polygons, text on the canvas

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References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

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